



Flood Probability Classifier — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

Version 1.0 — 18-March-2026

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1 Test Results — Flood Probability Classifier (MKM-FC-001)

Tests run on **18 March 2026 at 08:10:42**.

Table 1: Test Results Summary — Flood Probability Classifier (MKM-FC-001)

Total Tests	34
Passed	34
Failed	0
Skipped	0
Duration	0.34s

Table 2: Detailed Test Results — Flood Probability Classifier

Test Description	Acceptance Criteria	Result	Time
Each storm has peak_level_m.	'peak_level_m' in s; s['peak_level_m'] > 0	Pass	0.001s
Storms have required identification fields.	'storm_id' in s; 'name' in s (+3 more)	Pass	0.002s
Returns 404 when stress_storms.json missing.	resp.status_code == 404	Pass	0.001s
Missing gauge_id returns total count (used by startup preloader).	resp.status_code == 200; data['status'] == 'success' (+2 more)	Pass	0.002s
Returns empty list for gauge with no storms.	resp.status_code == 200; data['storms'] == []	Pass	0.002s
Returns storm list for specified gauge.	resp.status_code == 200; data['status'] == 'success' (+2 more)	Pass	0.002s
Storms sorted by peak level descending (worst first).	peaks == sorted(peaks, reverse=True)	Pass	0.002s
Each gauge has gauge_id, gauge_name, lon, lat, trade_count.	'gauge_id' in g; 'gauge_name' in g (+3 more)	Pass	0.013s
Gauges with trades show non-zero trade_count.	sum(counts) > 0	Pass	0.017s
Returns gauge list with trade counts.	resp.status_code == 200; data['status'] == 'success' (+2 more)	Pass	0.014s
Gauges sorted west to east (ascending longitude).	lons == sorted(lons)	Pass	0.013s
Each hour has water_level, p_flood, portfolio fields.	'hour' in h; 'water_level' in h (+4 more)	Pass	0.023s

Test Description	Acceptance Criteria	Result	Time
Non-existent storm returns 404.	<code>resp.status_code == 404</code>	Pass	0.003s
Severe storm triggers knock-out (P(flood)=100%).	<code>data['hourly'][first_breach]['p_flood'] == 1.0; h['p_flood'] is None</code>	Pass	0.022s
Missing gauge_id or storm_id returns 400.	<code>resp.status_code == 400</code>	Pass	0.000s
No JSON body returns error.	<code>resp.status_code in (400, 500)</code>	Pass	0.002s
Response includes portfolio-level stress summary.	<code>'num_trades' in summary;</code> <code>'total_notional' in summary</code> (+5 more)	Pass	0.023s
Response includes probability surface grid.	<code>'probability_surface' in data;</code> <code>'water_levels' in surface</code> (+7 more)	Pass	0.024s
POST returns 60-hour forecast array.	<code>resp.status_code == 200;</code> <code>data['status'] == 'success'</code> (+1 more)	Pass	0.021s
Response includes per-trade summary.	<code>'trades' in data; len(data['trades']) == 2</code> (+4 more)	Pass	0.021s
Surface water levels cap at severe — no rows above severe.	<code>max(surface['water_levels']) <= severe</code>	Pass	0.041s
Surface must cover full 168h storm horizon regardless of KO.	<code>max(surface['hours']) >= 164</code>	Pass	0.033s
Empty gauge_id → 400 (line 217).	<code>resp.status_code == 400</code>	Pass	0.001s
No open trades at gauge → 404 (line 250).	<code>resp.status_code == 404</code>	Pass	0.002s
Null JSON body → 400 (line 205).	<code>resp.status_code == 400</code>	Pass	0.000s
Predictor None (no model dir) → 404.	<code>resp.status_code == 404</code>	Pass	0.019s
Gaugets file present → hydrograph built from real readings (lines 258-279).	<code>resp.status_code == 200;</code> <code>data['status'] == 'success'</code>	Pass	0.032s
<code>_get_predictor</code> returns cached value without re-importing.	<code>result is sentinel</code>	Pass	0.000s
<code>_get_predictor</code> returns None when both stressm and legacy stress dirs are empty.	<code>pred is None</code>	Pass	0.000s
stress/run returns 500 on engine error (lines 302-306).	<code>resp.status_code == 500</code>	Pass	0.003s
stress/gauges returns 500 when gauge loader raises (lines 122-124).	<code>resp.status_code == 500</code>	Pass	0.001s
stress/storms returns 500 when storm loader raises (lines 185-187).	<code>resp.status_code == 500</code>	Pass	0.001s

Test Description	Acceptance Criteria	Result	Time
peak_position=0 uses decay-only path.	<code>len(levels) == 10; all(isinstance(v, float) for v in levels)</code>	Pass	0.000s
remaining=0 uses frac=1.0 path.	<code>len(levels) == 5</code>	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
04-Mar-2026	1.0	Initial test suite (15 tests)	D. Kelly
05-Mar-2026	1.1	7 test(s) added (total: 22)	D. Kelly
09-Mar-2026	1.2	56 test(s) added; 22 test(s) removed (total: 56)	D. Kelly
09-Mar-2026	1.3	4 test(s) added; 4 test(s) removed (total: 56)	D. Kelly
10-Mar-2026	1.4	12 test(s) added (total: 68)	D. Kelly
10-Mar-2026	1.5	3 test(s) added (total: 71)	D. Kelly
12-Mar-2026	1.6	30 test(s) added (total: 101)	D. Kelly
16-Mar-2026	1.7	37 test(s) removed (total: 64)	D. Kelly
17-Mar-2026	1.8	1 test(s) added; 1 test(s) removed (total: 64)	D. Kelly
18-Mar-2026	1.9	30 test(s) removed (total: 34)	D. Kelly